Mar. 19

going through proof in HW 8

- **Cauchy's Theorem** Let G be a finite group and let p be a prime dividing $|G|$. Then G has an element of order p.
  - Consequence: Say $a \in G$ has order p, $\langle a \rangle$ is a subgroup of G of order p

partial converse to Lagrange

- Group Actions & Orbit-Stabilizer Theorem: used in proof of Cauchy in HW

a Mersenne prime

Application of Cauchy Let G be an Abelian group of order $2 \cdot (2^{1032471} - 1)$. Claim: G is cyclic.

call this J

Proof

By Cauchy, $\exists x \in G$ with $|x| = J$ and $y \in G$ with $|y| = 2$. We see $|xy| = 2J$. Thus $G = \langle xy \rangle$.

textbook calls these factor groups

Chapter 9: Normal subgroups & Quotient Groups

- Idea: We want to generalize the idea behind going from $\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z}$

- **Def** A subgroup H of a group G is called normal if $aH = Ha$ for all $a \in G$. We denote this $\underbrace{H \trianglelefteq G}_{H \text{ is normal in } G}$

  i.e $\{aH : h \in H\} = \{Ha : h \in H\}$ for all $a \in G$

Ex $H = \langle 4 \rangle = 4\mathbb{Z}$ in $G = \mathbb{Z}$

$\mathbb{Z}$ is an Abelian group, so $a + H = \{a + 4k : k \in \mathbb{Z}\} = \{4k + a : k \in \mathbb{Z}\} = H + a$

- **Prop** If G is Abelian and $H \leq G$, then $H \trianglelefteq G$

  Proof sketch $ah = ha$ since elements of G commute

- Examples of Normal Subgroups
  - $H = \langle (123) \rangle \leq S_3$

$H = eH = (123)H = (132)H$

$\parallel$

$H = He = H(123) = H(132)$

b/c cosets are either equal or disjoint

Lagrange: $|S_3| = [S_3 : H] \cdot |H|$

$6 = [S_3 : H] \cdot 3 \Rightarrow [S_3 : H] = 2$

$\{(12), (23), (13)\}$

$(12)H = (23)H = (13)H$

$\parallel$

$H(12) = H(23) = H(13)$

$\{(12), (123)(12), (132)(12)\}$

$\{(12), (13), (32)\}$

★Important note: either $aH \cap bH = \emptyset$ or $aH = bH$ for two left or two right cosets, but doesn't necessarily hold for one left one right. We could have $aH \cap Hb \neq \emptyset$ and $aH \neq Hb$

Therefore, we've shown $aH = Ha$ for all $a \in S_3$. Thus $H = \langle (123) \rangle \trianglelefteq S_3$.

- **Thm. 9.1 (Criterion for normality)** $H \trianglelefteq G \iff aHa^{-1} \subseteq H$ for all $a \in G$

$aHa^{-1} = \{aha^{-1} : h \in H\}$ Sometimes called conjugation by a

Ex. $H = \langle (123) \rangle$, $a = (12) = a^{-1}$

$aHa^{-1} = (12)H(12) = \{(12)e(12), (12)(123)(12), (12)(132)(12)\} = \{e, (132), (123)\} = H$

Notice elements are permuted but set is unchanged

Proof

($\Rightarrow$) Suppose $H \trianglelefteq G$. Let $x \in aHa^{-1}$, so $x = aha^{-1}$ for some $h \in H$. Since $H \trianglelefteq G$, $aH = Ha$, so $ah = h'a$ for some $h' \in H$. Therefore, $x = aha^{-1} = h'aa^{-1} = h' \in H$. So $aHa^{-1} \subseteq H$.

($\Leftarrow$) Suppose $aHa^{-1} \subseteq H$ for all $a \in G$. Let aH be an arbitrary left coset, and let $ah \in aH$ be an arbitrary element. Since $aHa^{-1} \subseteq H$, we see $aha^{-1} = h'$ for some $h' \in H$. Thus $ah = h'a \in Ha$. Hence $aH \subseteq Ha$. Since $|aH| = |Ha|$, we see $aH = Ha$, so $H \trianglelefteq G$. $\square$

- **Theorem 9.2 (Hölder's Concept of Quotient Groups)**

Let $N \trianglelefteq G$, then the set $G/N = \{aN : a \in G\}$ is a group under the operation $(aN)(bN) = abN$

The Quotient of G by N or "G mod N"

We are forming a group of the cosets of a normal subgroup

Caution: G/N is <u>not</u> a subgroup of G

Proof

first we want to show our binary operation is well-defined. (We don't want issues with different coset representatives.

Suppose $aN = a'N$ and $bN = b'N$. We want $abN = a'b'N$. Since $aN = a'N$, we have $ae = a = a'n_1$ for some $n_1 \in N$. Similarly, $b = b'n_2$ for some $n_2 \in N$. Thus $abN = a'n_1b'n_2N$.

Now, since $N \trianglelefteq G$, we see $b'N = Nb'$, so $n_1b' = b'n_3$ for some $n_3 \in N$. Thus $a'n_1b'n_2N = a'b'n_3n_2N = a'b'N$.

associativity: from group

identity: eH

inverses: $(aH)^{-1} = a^{-1}H$